



Newer Macbook 'Pro'

The Macbook line-up hasn't updated in over a year, but a new 'pro' is on its way
<http://dgit.in/MbPro16>



Tesla in flames

Check out how a Tesla Model S burst into flames during a test drive
<http://dgit.in/TslBurn>

Feature

Innovating on the basic process of yarn creation, making Jacquard yarn involves using conductive metal alloys along with traditional thread yarns like cotton, silk or polyester. This makes the yarn strong enough to be woven on any industrial loom. In fact, this is the theoretical principle using which many fashion companies are trying to implement smart fabrics.

Simply being able to conduct electricity wouldn't make clothing, or anything else, smart. Conductors and circuits no larger than a shirt button can be used to harness the connectivity provided by these smart fabrics. These devices in turn can sense a number of inputs, including touch, temperature, vibrations etc. Using wireless transmission, these interactions can be connected to external devices, with the most suitable candidate being smartphones, due to their constant use.

Now, when it comes to the components involved in the circuit, there are actually quite a few options. Embedded electronics have been miniature in size for a long while and integrated circuits are increasing density while reducing size every year. In all of this, perhaps the most significant example of a device created for the purpose of smart clothing is the Arduino Lilypad.

The Arduino open source platform is very popular for the purpose of prototyping. One of its speciality boards, the Lilypad, is meant for implementation in smart fabric and wearable projects. It consists of several components depending on the particular Lilypad model that you chose. There is a USB specific model as well as a Simple model. With a flash memory of 16KB and processing clock speed of 8MHz, it can be configured for specific use cases using its 14 digital I/O pins. It can be programmed using the Arduino Software (IDE).

As for implementations, Google's Jacquard project has been implemented in partnership with Levi's in the form of the Levi's Commuter Trucker jacket, which is essentially meant to help you navigate your mobile screen while navigating your city streets on a bike.

The connectivity allows you to perform simple gestures using swipes and taps over a tag embedded into the garment, which in turn seamlessly connects to your smartphone placed in the jacket's pocket.

Fashion forward

Beyond performance enhancements and smart-connectivity, there exist quite a few growing uses of smart fabrics. These incorporate smart features that deal with more intrinsic and traditional uses of clothing. One such primary use is keeping us warm, and smart clothing manufacturers have taken that up as well. Australian

Our Contribution

Be it in the pursuit of technical objectives, or a reaction to various assault cases across the country, students in India are coming up with novel ways of giving people ways to protect themselves via their clothing. A few students from Chennai's SRM university got together to develop an anti-rape underwear, which essentially looks like a nightgown but is capable of giving an assaulter a 3800kV shock at the press of a button, along with sending an alert to the police. A couple of students from NIFT have also developed an anti-molestation jacket that achieves the same purpose. A major commonality



An example implementation of the Jacquard yarn for controlling a music player

Wool Innovators, a garment company from down south has its eyes set on creating socks that keep your toes at a specific temperature, using a power source and a conductive fabric used in the socks.

Another common problem that smart fabrics aim to solve is odour. This is mainly being done by treating the fabric with certain chemicals or elements, with silver being in the forefront. According to experts, treating the garment with silver takes care of the odour permanently. And microencapsulation is opening up a huge number of possibilities like including aloe vera, vitamins or insect repellents into the fabric. This could be very useful for medical purposes, by including healing agents as well.

between the two projects has been the target of keeping the end result affordable for everyone.

In Conclusion

While you cannot get your own touch sensitive biker jacket right now, especially in India, smart fabrics have been around in certain forms and are being innovated upon everyday. Market predictions say that smart clothing will overtake the demand for other wearables quite soon. With this significant demand it is undoubtedly going to be an exciting field with things like the Jacquard yarn soon becoming commonplace. While ethical and security concerns will still remain, there is no denying that being in fashion is soon going to be a smart choice! 